

ECON 609 - Problem Set 1

Fall 2022

Be sure to include all the material that shows your work and to present your results in a way that is easy to understand.

Dynare

Read carefully the note written by Sims on Dynare. Write the .mod file that runs on Matlab. Generate the figures including the simulated data and the impulse responses.

Kalman Filter

Consider the following linear state-space representation

$$\begin{aligned}y_t &= \Psi s_t + u_t \\s_t &= \Phi s_{t-1} + \epsilon_t\end{aligned}$$

where $\mathbb{E}[u_t u_t'] = \Sigma_u$ and $\mathbb{E}[\epsilon_t \epsilon_t'] = \Sigma_\epsilon$. Let the dimension of y_t be 1×1 and s_t be 2×1 . Consider the following parameterization

$$\Psi = \begin{bmatrix} 1 & 0.9 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.5 \end{bmatrix}, \quad \Sigma_u = 0.1^2, \quad \Sigma_\epsilon = \begin{bmatrix} 0.3^2 & 0 \\ 0 & 0.25^2 \end{bmatrix}.$$

(1) Plot the Data

Generate $T=50$ observations from this state-space model and plot y_t and s_t . You can simply set $s_0 = 0$.

(2) Kalman Filter

Write a code to run the Kalman filter and store the mean and variance of $p(s_t|Y_{1:t})$ as well as the log likelihood increments $\log p(y_t|Y_{1:t-1})$.

(3) Performance of Kalman Filter

To make sure that the filter works properly, plot $\mathbb{E}[s_t|Y_{1:t}]$ as well as 90% credible sets based on $p(s_t|Y_{1:t})$. Overlay the true states s_t on top.

Metropolis Hastings Sampler

The Matlab program Metropolis.m uses the RWMH algorithm to generate draws from a bimodal distribution. Read through the program and run it with the default setting.

(1) Change in Parameters

Vary the parameters μ_1, μ_2, Σ , and p to explore the performance of the Metropolis Hastings algorithm. What happens to the accuracy of the numerical approximation?

(2) Change in Other Settings

Vary the number of draws and the variance-covariance matrix of the proposal density. What happens to the accuracy of the numerical approximation?

Value Function Iteration

The remaining questions will be based on the following model. There is a representative household with preferences:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} 0.96^t \left(\log c_t - \frac{l_t^2}{2} \right)$$

The household consumes, saves, and works, with a budget constraint

$$c_t + i_t = w_t l_t + r_t k_t$$

There is a production function

$$c_t + i_t = e^{z_t} k_t^{\alpha_t} l_t^{1-\alpha_t}$$

with a law of motion for capital

$$k_{t+1} = 0.9k_t + i_t$$

where $i_t \geq 0$ and a technology level z_t that follows a Markov Chain that takes values in

$$z_t \in \{-0.0673, -0.033, 0, 0.0336, 0.0673\}$$

with transition matrix

$$\begin{bmatrix} 0.9727 & 0.0273 & 0 & 0 & 0 \\ 0.0041 & 0.9806 & 0.0153 & 0 & 0 \\ 0 & 0.0082 & 0.9836 & 0.0082 & 0 \\ 0 & 0 & 0.0153 & 0.9806 & 0.0041 \\ 0 & 0 & 0 & 0.0273 & 0.9727 \end{bmatrix}$$

and $\alpha_t \in \{0.25, 0.3, 0.35\}$ with transition matrix

$$\begin{bmatrix} 0.9 & 0.07 & 0.03 \\ 0.05 & 0.9 & 0.05 \\ 0.03 & 0.07 & 0.9 \end{bmatrix}$$

You can think about α_t as the reduced form of some technological change that modifies the elasticity of output with respect to capital.

(1) Social Planner

Find the associated social planner's problem to this model and write it recursively.

(2) Steady State

Compute the deterministic steady state of the model when $z_{ss} = 0$ and $\alpha_{ss} = 0.3$.

(3) Value Function Iteration with a Fixed Grid

Fix a grid of 250 points of capital, centered around k_{ss} with a coverage of $\pm 30\%$ of k_{ss} and equally spaced. Iterate on the value function implied by the social planner's problem using linear interpolation until the change in the sup norm between the iterations is less than 10^{-5} . Compute the policy function.

(4) Discussion

Provide the discussion in terms of i) accuracy, ii) computing time, and iii) complexity of implementation of this solution method.